THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Is 4725 divisible by 9? 1

(a) Only if the last digit is even **(b)** No

(c) Yes **(d)** Only if the number is prime

2. Is 2461 divisible by 3? 1

(a) Only if the number is prime **(b)** Yes

(c) Only if the last digit is even **(d)** No

3. Is 1331 divisible by 11? 1

(a) Yes **(b)** Only if it is odd

(c) No **(d)** Only if it ends in 1

4. The smallest digit that replaces the blank in 36_2 so that the number is divisible by 9 is: 1

(a) 9 **(b)** 2 **(c)** 7 **(d)** 8

5. The LCM of 12 and 18 is: 1

(a) 36 **(b)** 6 **(c)** 48 **(d)** 216

6. The LCM of 16 and 24 is: 1

(a) 384 **(b)** 8 **(c)** 48 **(d)** 64

7. The smallest number divisible by 2, 3 and 5 is: 1

(a) 10 **(b)** 60 **(c)** 30 **(d)** 15

8. The product of an even and an odd number is: 1

(a) a square **(b)** odd **(c)** prime **(d)** even

9. The number 1 is: 1

(a) neither prime nor composite **(b)** prime

(c) composite **(d)** even

10. A number ending in 00 is always divisible by: 1

(a) 100 **(b)** 3 **(c)** 7 **(d)** 11

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the divisibility test for 3. 2

12. State the divisibility test for 11. 2

13. Is 5148 divisible by 4? Explain. 2

14. Check whether 7392 is divisible by 8. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find all digits that can replace the blank in 5_72 to make it divisible by 3. 3

16. Find all digits that can replace the blank in 3_47 to make it divisible by 9. 3

17. Find all digits that can replace the blank in 8_16 to make it divisible by 9. 3

OR

Find all digits that can replace the blank in 7_29 to make it divisible by 3. 3

18. Find all digits that can replace the blank in 4_51 to make it divisible by 9. 3

OR

Find all digits that can replace the blank in 9_36 to make it divisible by 3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) State the divisibility tests for 2, 4, 5 and 8.
(ii) Test 3,15,000 and 59,32,816 for these. 5

OR

(i) Find the HCF and LCM of 84 and 126 by prime factorisation. 5
(ii) Verify HCF × LCM = product.

20. **Case study:** Three traffic lights change every 40 s, 60 s and 90 s respectively and all turn green together at 9:00 a.m. 5
(i) After how many seconds will they next change together?
(ii) At what time?
(iii) How many times will they change together between 9:00 and 10:00, counting both 9:00 and 10:00?

OR

(i) Solve the cryptarithm $AB + BA = 121$ for digits A, B. 5
(ii) Show that the sum of a two-digit number and its reverse is always a multiple of 11.

END OF QUESTION PAPER — Total: 40 Marks

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